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Circuits 1 Lab #04
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Lab Report 04

BJT Characterization

Purpose: To practice and understand the practical uses of a BJT and understand their characteristics.

Materials:
1 4.7 kiloOhlm resistor
1 2N3904 BJT

Experient 1
BJT testing using multimeter

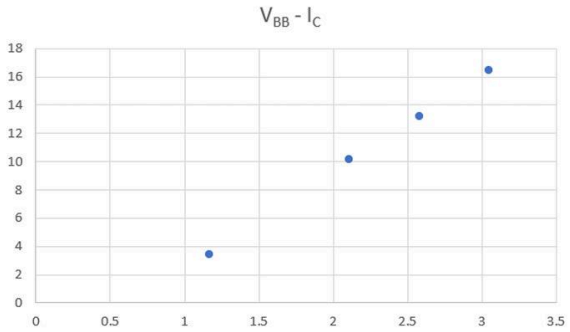
Data collected:

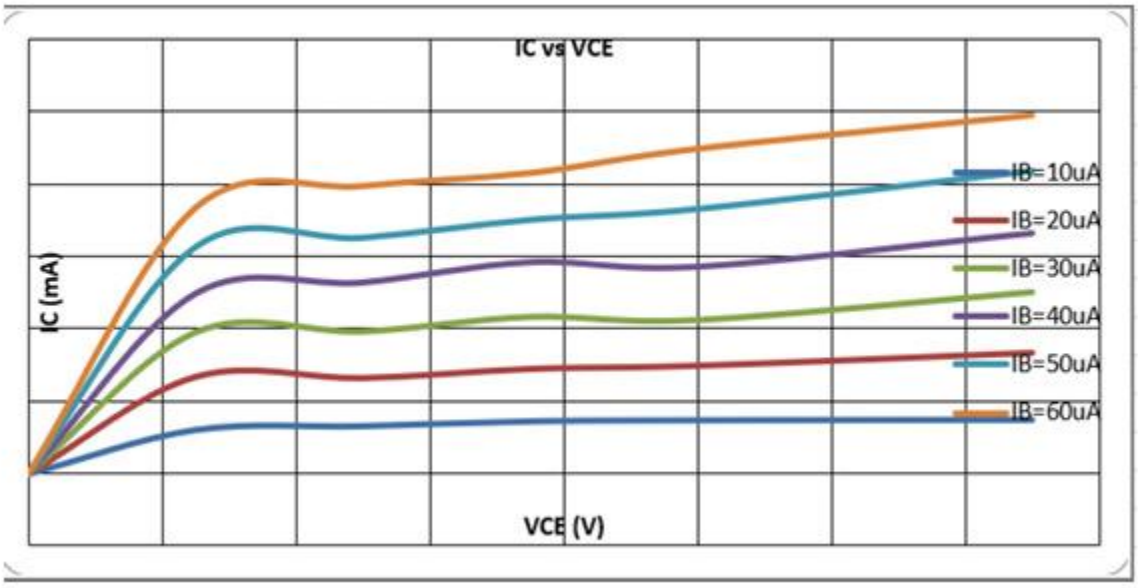
I_B [µA]	V_{CE} [V]					
	0	2.5	5	7.5	10	15
	I_C [mA]					
10	-0.0033	3.01	3.27	3.6	3.67	3.7
20	-0.0038	6.65	6.57	7.20	7.42	8.3
30	-0.0039	9.74	9.8	10.8	10.59	12.5
40	-0.0042	12.43	13.2	14.6	14.29	16.6
50	-0.0043	15.65	16.29	17.53	18.28	20.9
60	-0.0039	18.39	19.9	20.80	22.474	24.8

Graphs of data

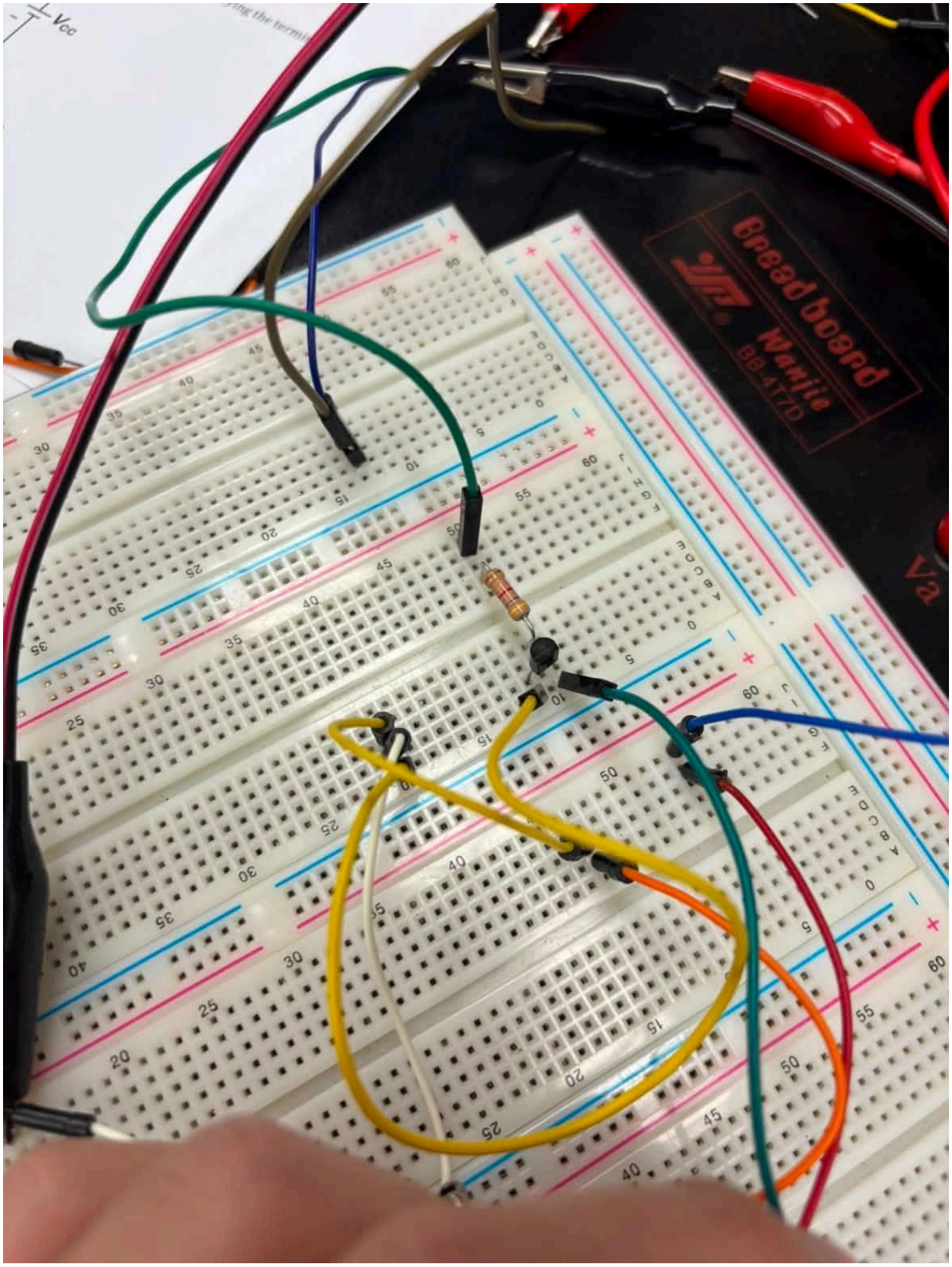
Data collected:

V_{BB} [V]	I_B [µA]	I_C [mA]	β
1.17	10	3.40	
			336
2.11	30	10.12	
			307
2.58	40	13.19	
			329
3.05	500	16.48	



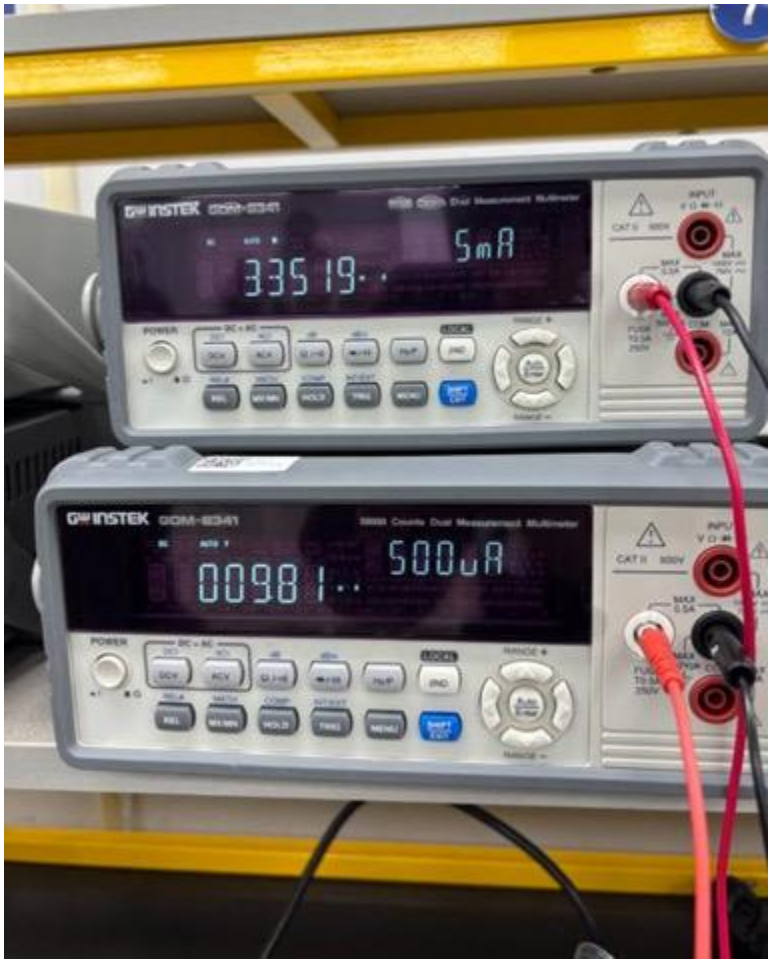


Circuit layout:

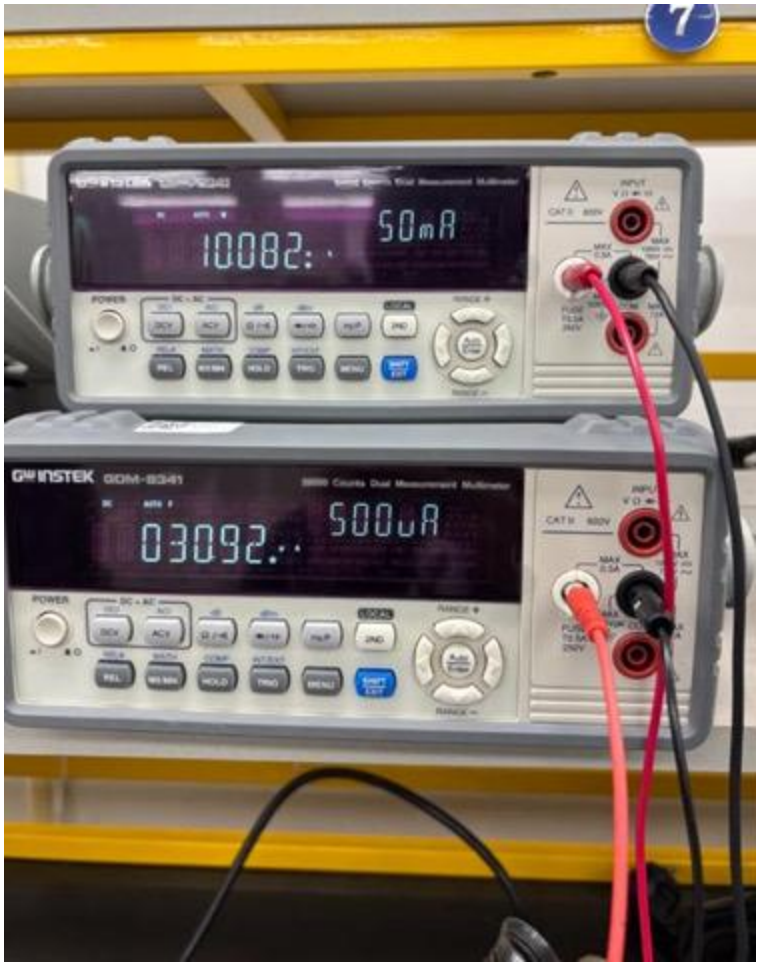

Vcc
Bread board
Wanlie
BB-4170

The results are shown below:

When Ic is 10 microamps:



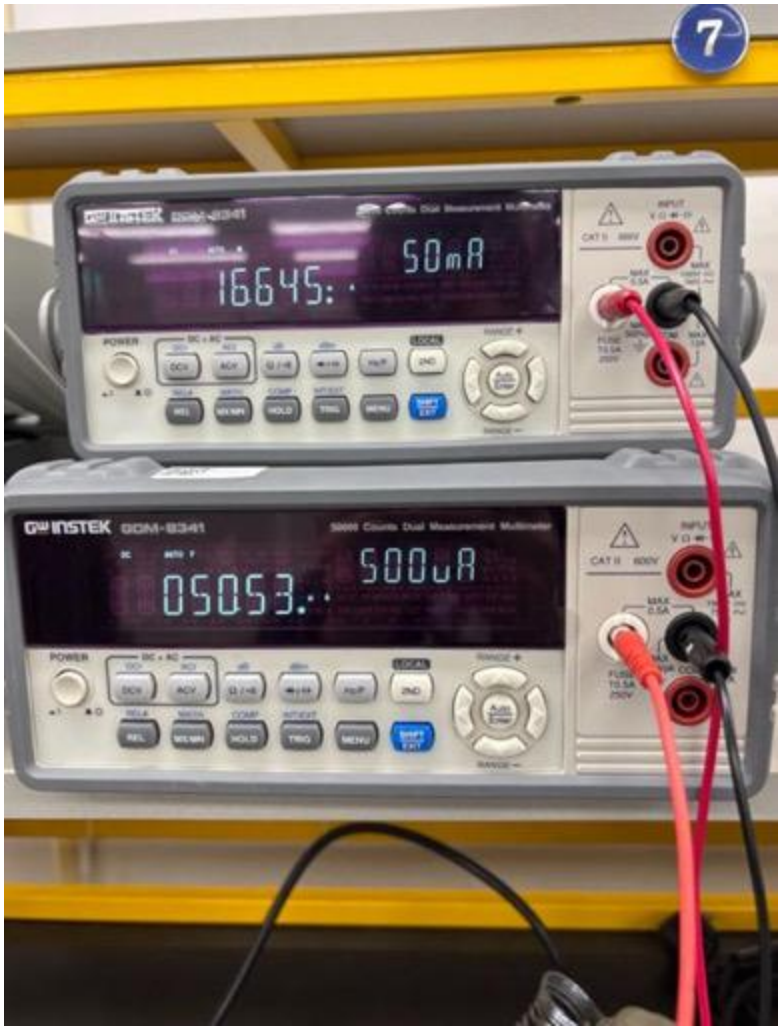
When Ic is 30 microamps:



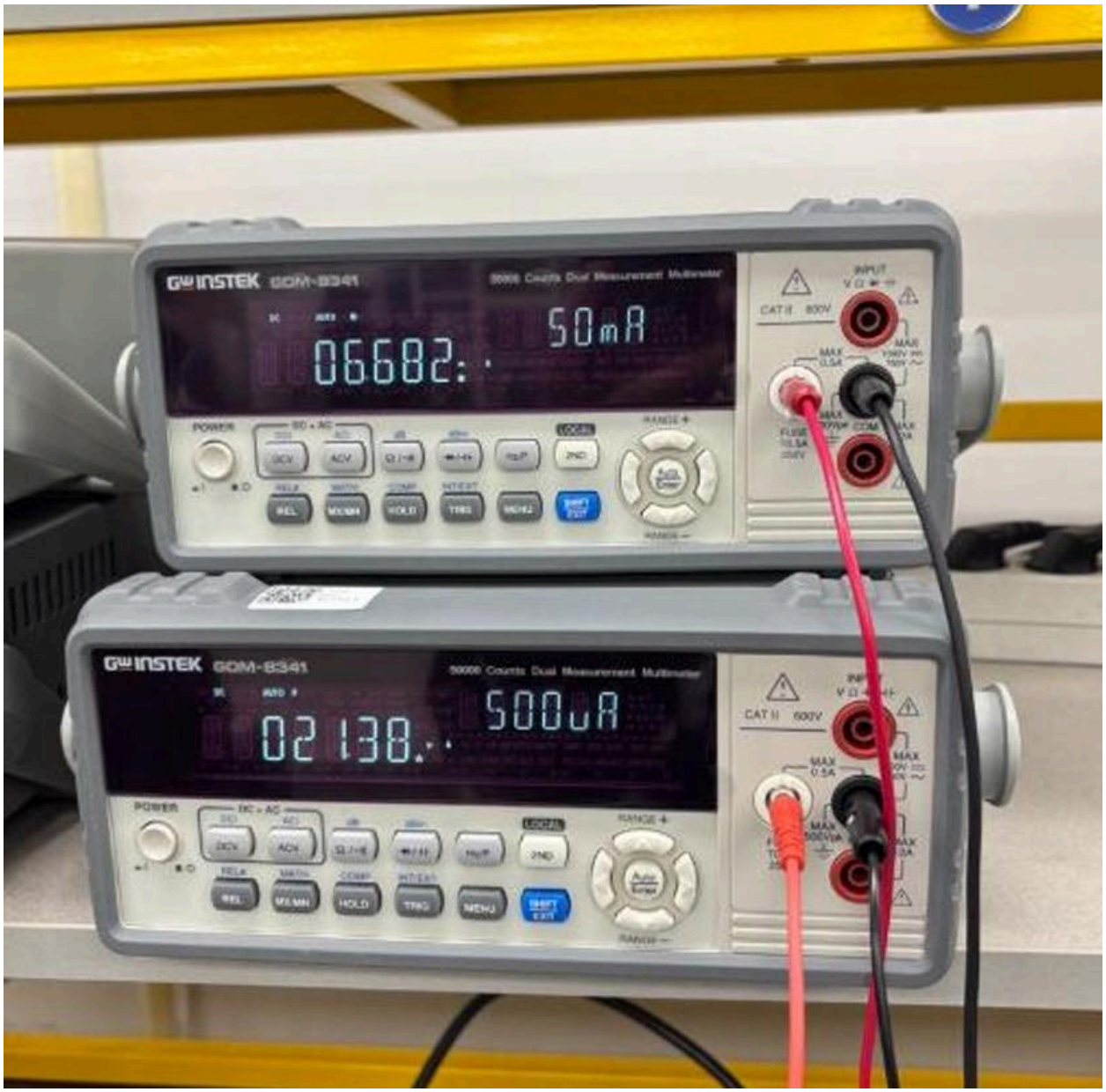
When Ic is 40 microamps:



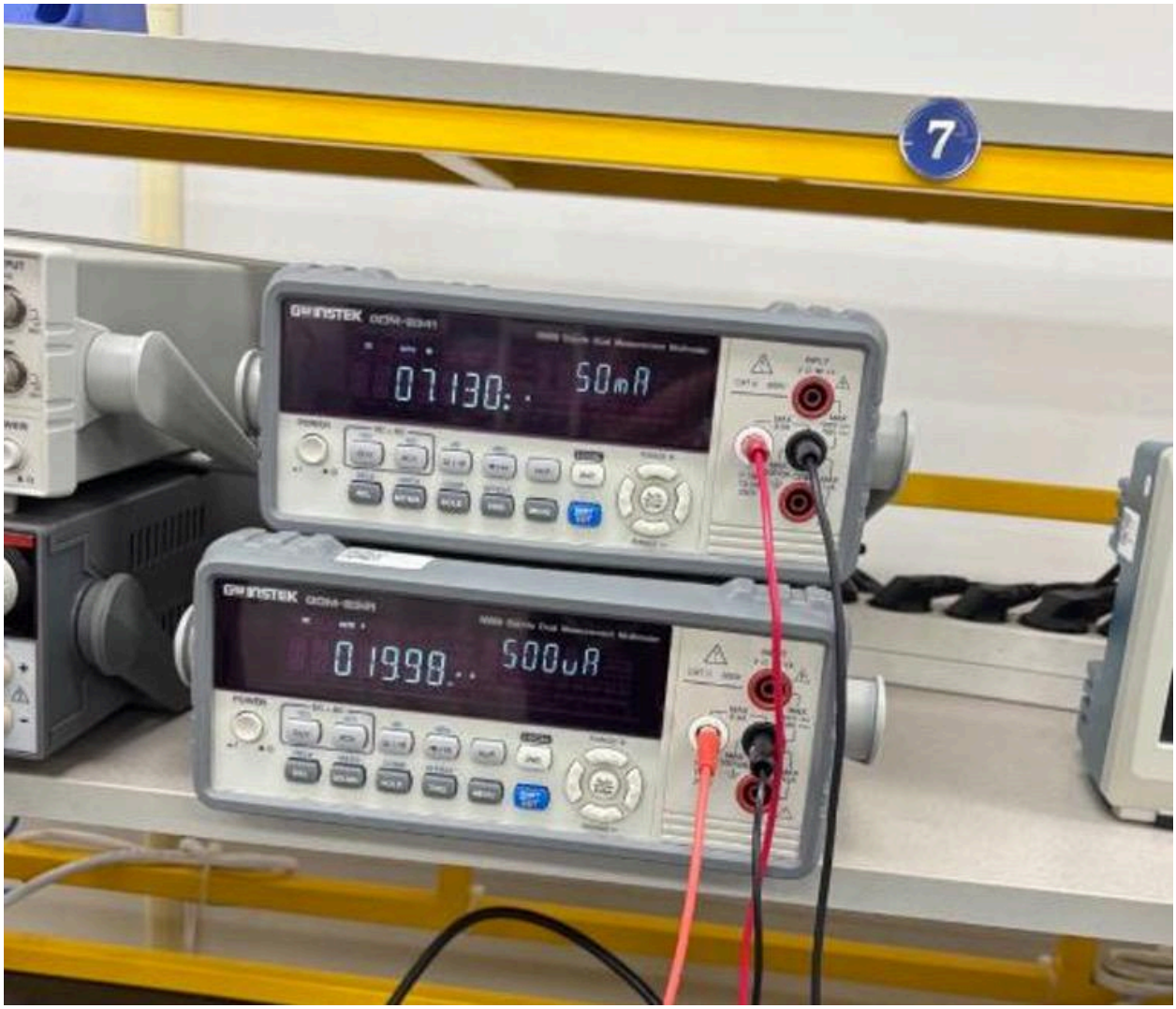
When Ic is 50 microamps:



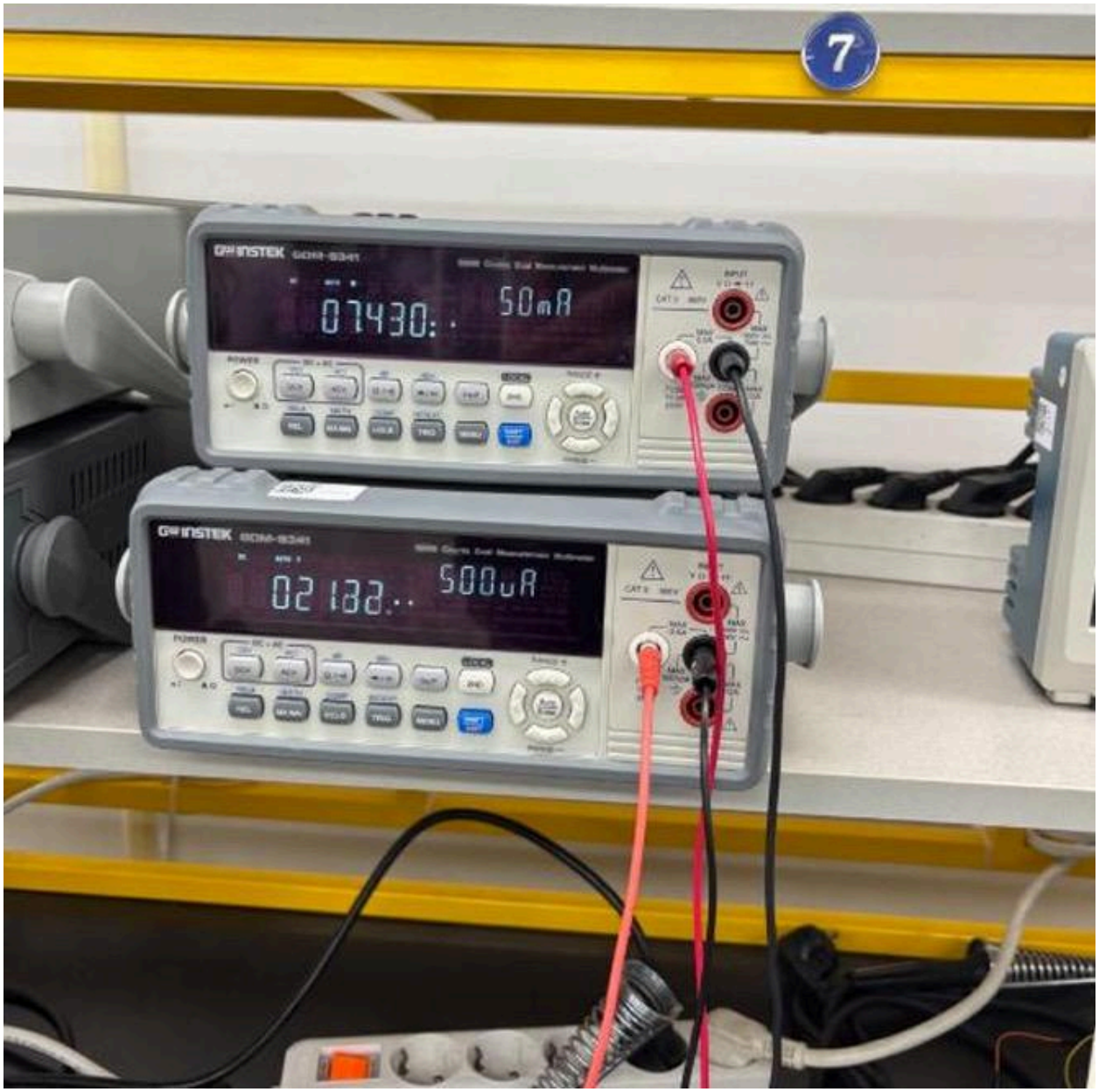
When the value of Ic when In is 20 micro amps and Vic is 5 volts:



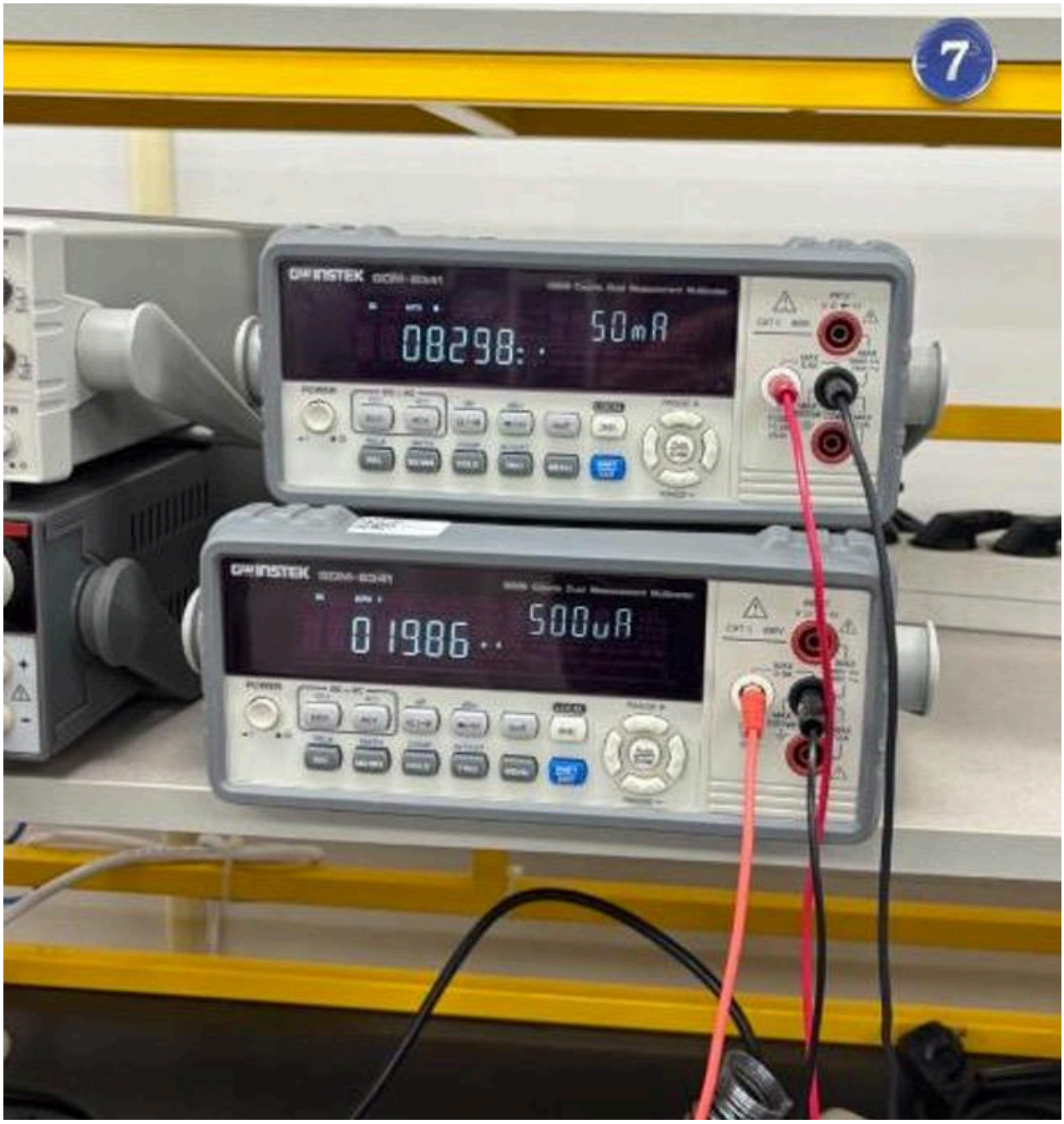
When the value of Ic when In is 20 micro amps and Vic is 7.5 volts:



When the value of Ic when In is 20 micro amps and Vic is 10 volts:

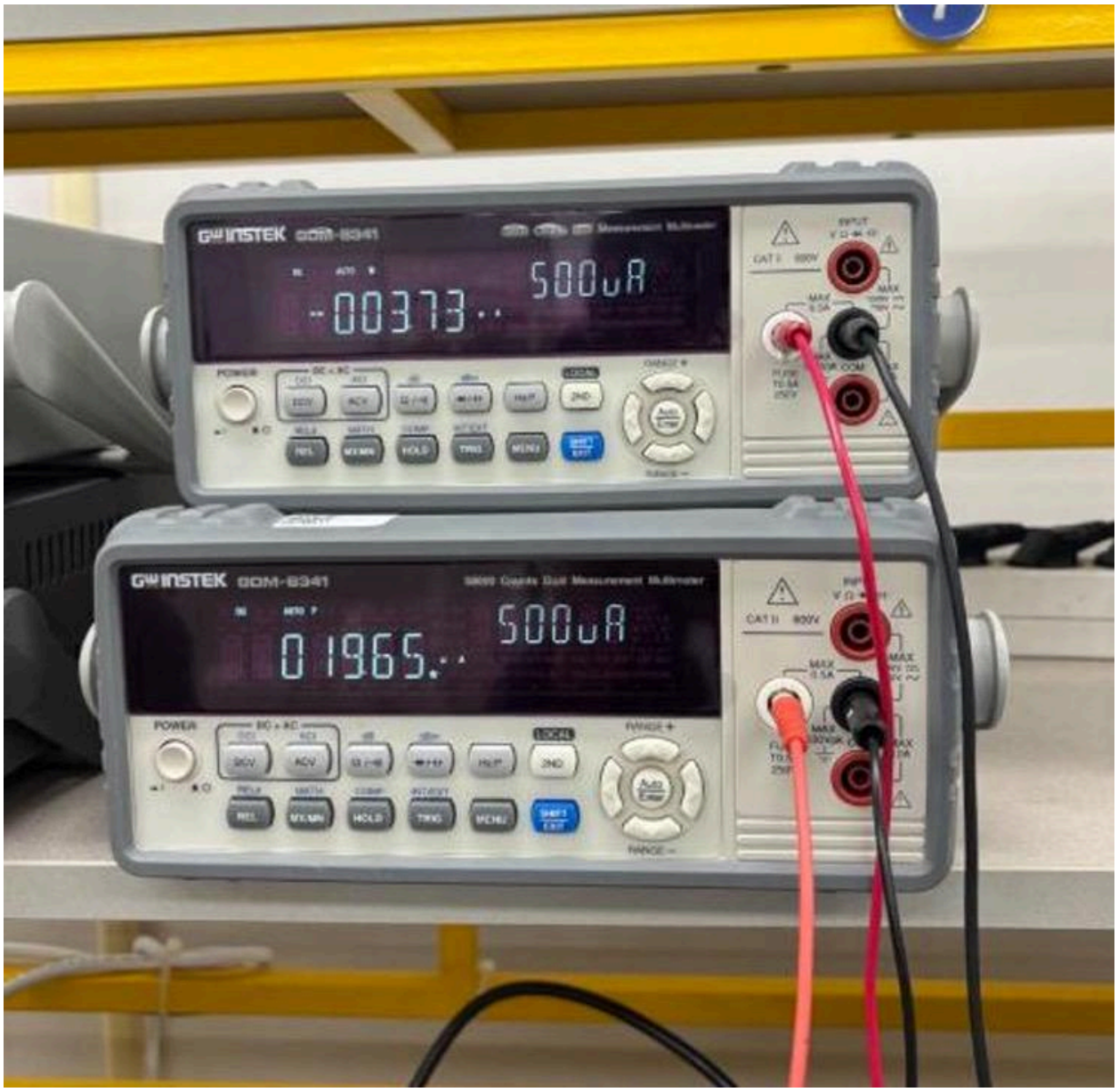


When the value of Ic when In is 20 micro amps and Vic is 15 volts:

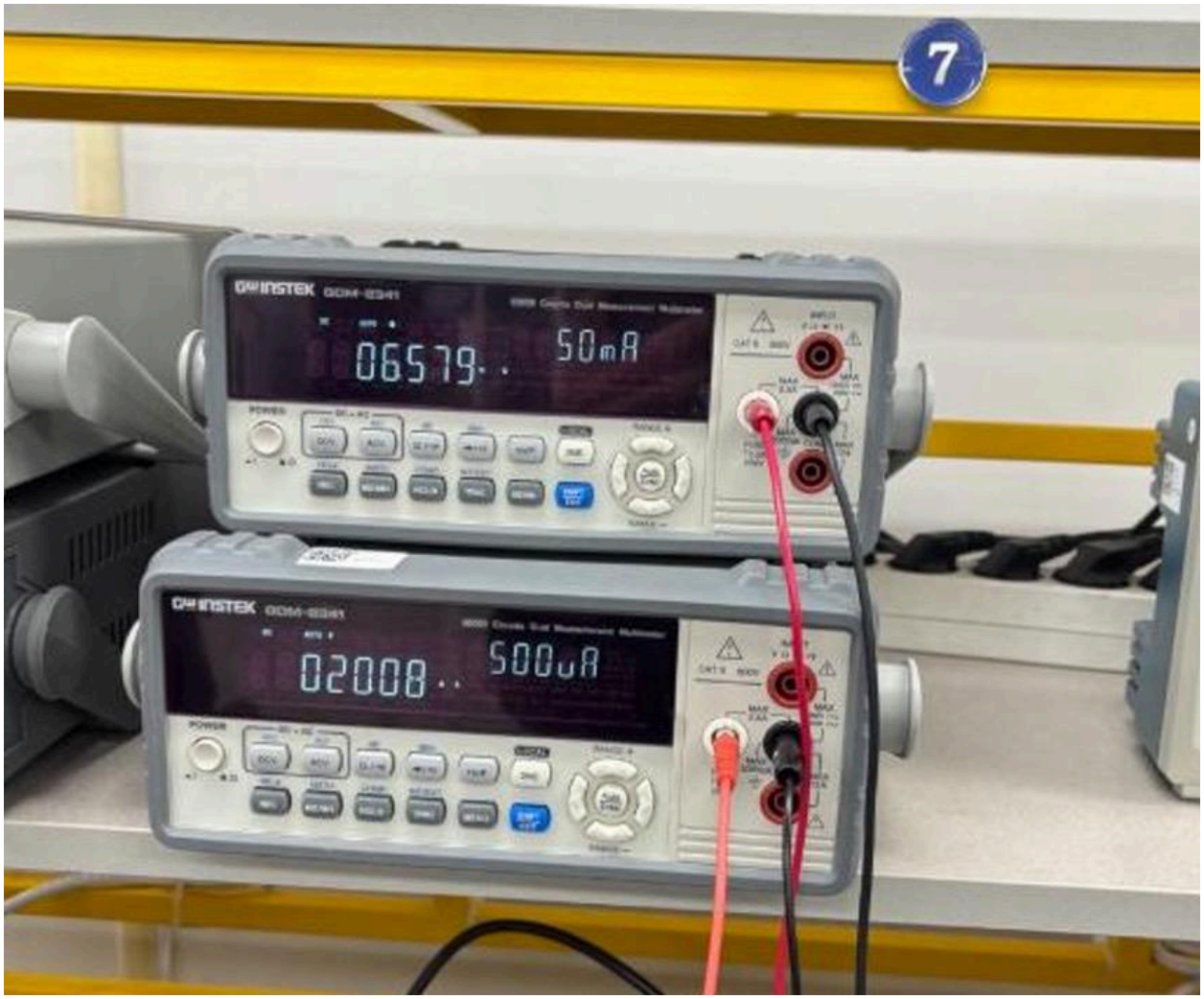


Results:

The value of Ic when Ib is 20 micro amps and Vc is 0 V

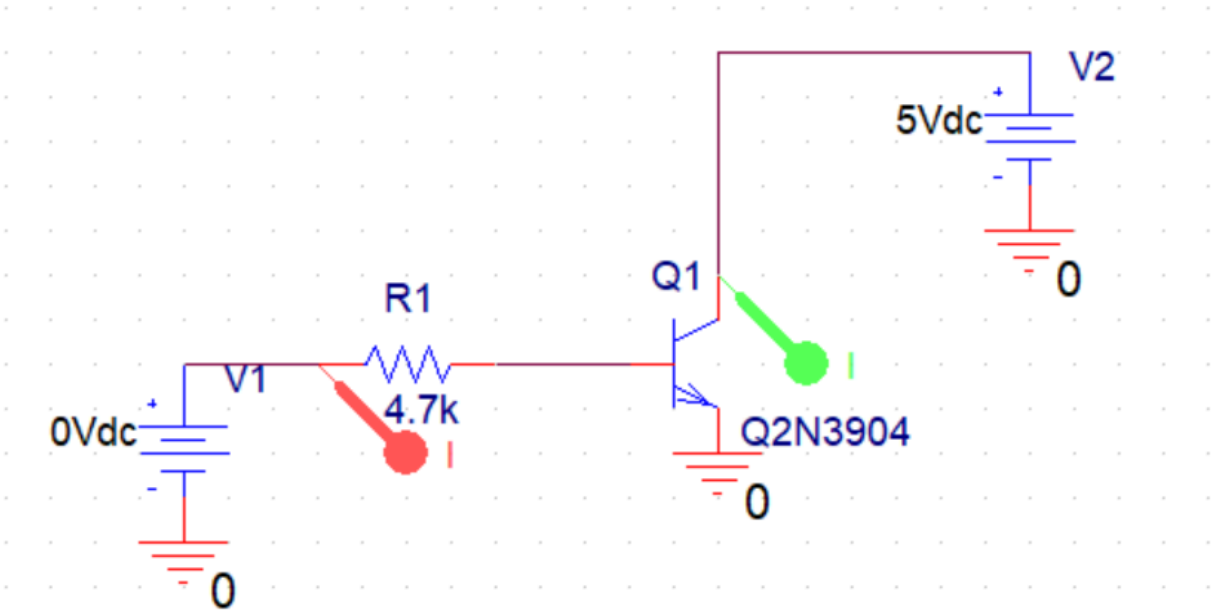


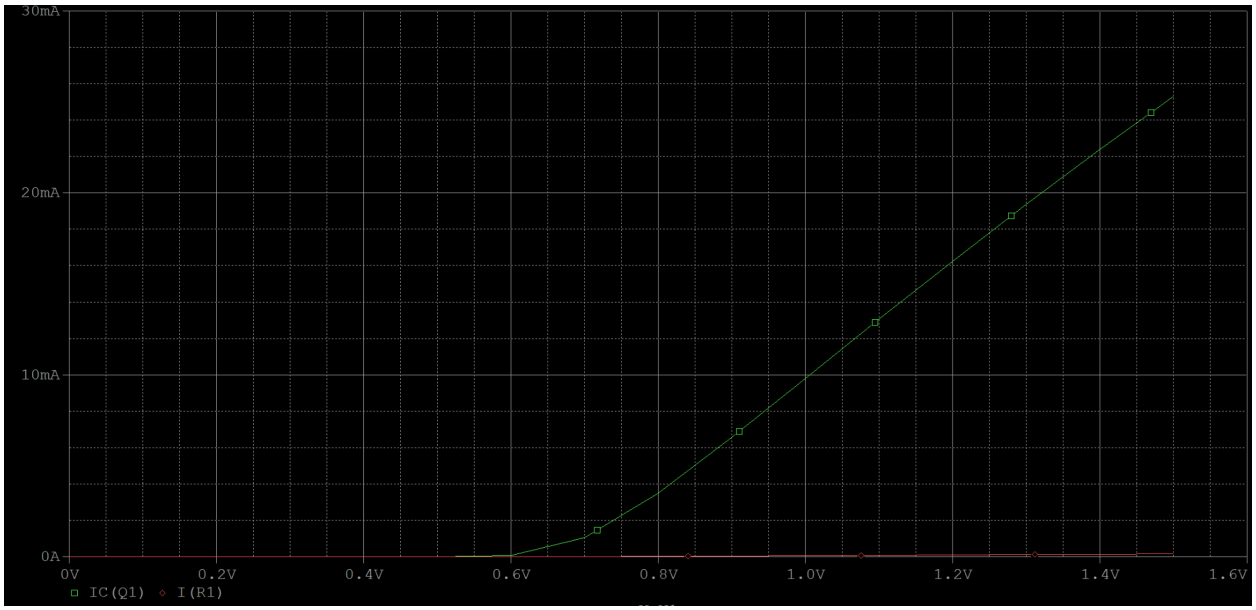
The value of Ic when Ib is 20 micro amps and Vc is 2.5 V

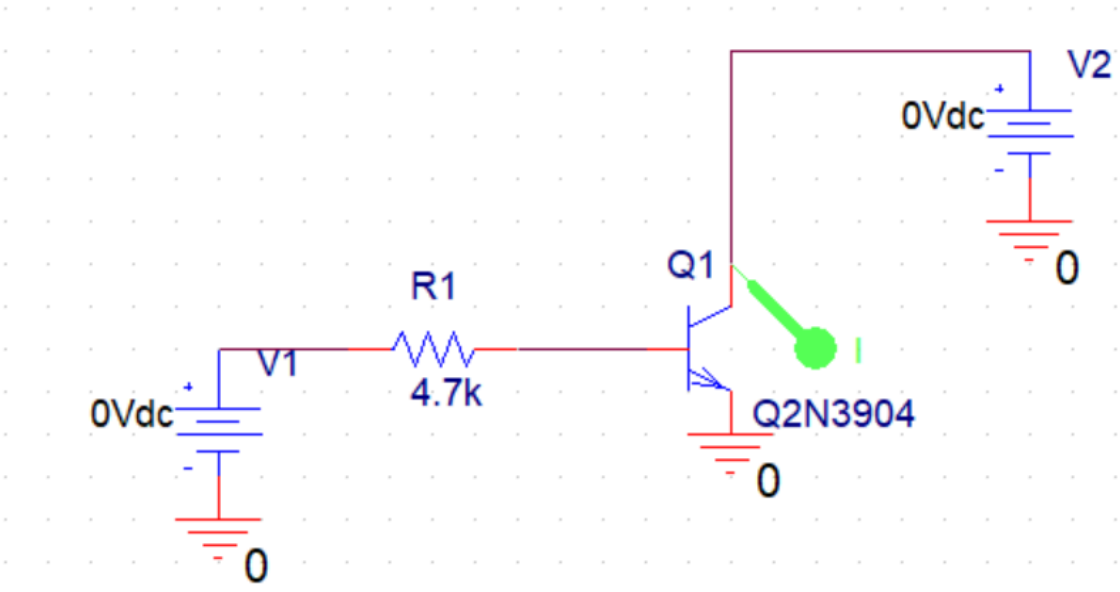


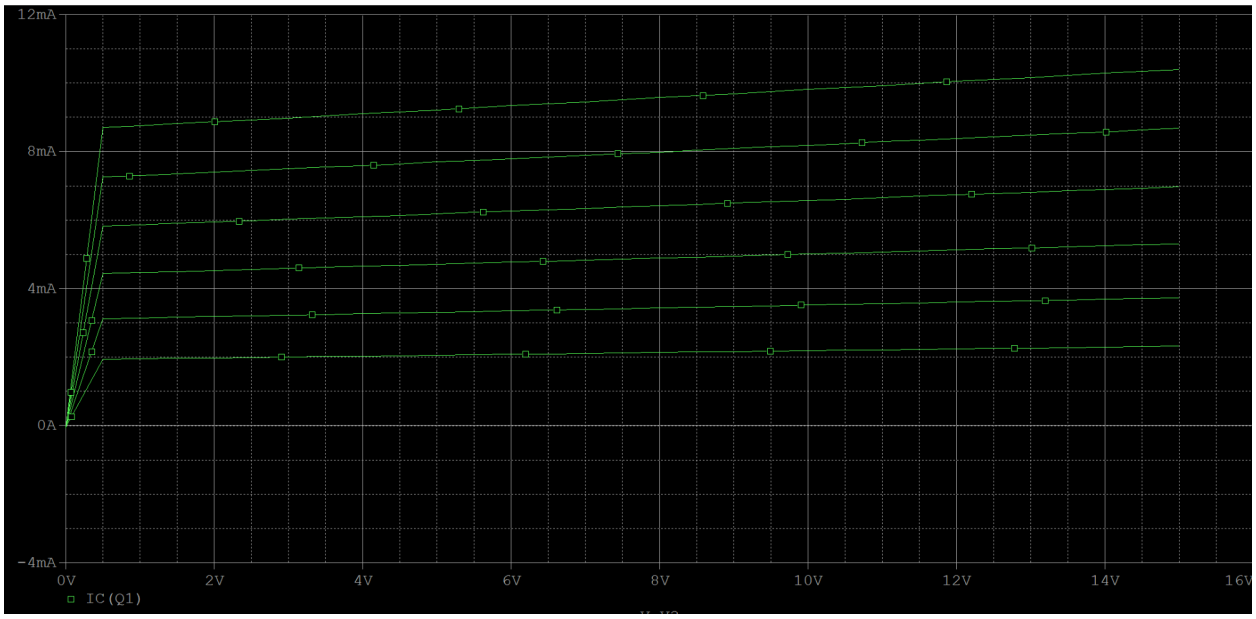
PSPICE:

Experiment 1:


V2
5Vdc
0
Q1
R1
V1
4.7k
0Vdc
Q2N3904
0
0


30mA
20mA
10mA
0A
0V
0.2V
0.4V
0.6V
0.8V
1.0V
1.2V
1.4V
1.6V
IC(Q1)
I(R1)


V2
0Vdc
0
R1
Q1
V1
4.7k
0Vdc
Q2N3904
0
0


12mA
8mA
4mA
0A
-4mA
0V
2V
4V
6V
8V
10V
12V
14V
16V
IC(Q1)